

Data Structure Lab 1 : Trees and Binary Trees

2024-2025

Topics

1. Create Position Interface
2. Create Tree Interface
3. Create AbstractTree Class.
4. Create BinaryTree Interface
5. Create AbstractBinaryTree Class
6. Create Linked Binary Tree Using Linked List structure (Nodes)
7. Implement Basic Methods of LinkedBinaryTree
 - addRoot(E e)
 - addLeft(Position<E> p ,E e)
 - addRight(Position<E> p ,E e)
 - set(Position<E> p ,E e)
 - remove(Position<E> p)

Homework

1. Implement the Array Based Binary Tree Data structure as it is described in chapter 8.

Data Structure Lab 10 : Trees and Binary Trees

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```
        }<public class Lab10<T
        ;private Object[] tree
        ;private int size

        } public Lab10(int capacity)
        ;tree = new Object[capacity]
        ;size = 0
        {

        } public void add(T element)
        } if (size == tree.length)
        ;()resize
        {
        ;tree[size] = element
        ;++size
        {

        } public T getLeftChild(int index)
        ;int leftChildIndex = 2 * index + 1
        } if (leftChildIndex < size)
        ;return (T) tree[leftChildIndex]
        {
        ;throw new IndexOutOfBoundsException("No left child")
        {

        } public T getRightChild(int index)
        ;int rightChildIndex = 2 * index + 2
        } if (rightChildIndex < size)
        ;return (T) tree[rightChildIndex]
        {
        ;throw new IndexOutOfBoundsException("No right child")
        {

        } public T getParent(int index)
        } if (index == 0)
        ;throw new IndexOutOfBoundsException("No parent for the root")
        {
        ;int parentIndex = (index - 1) / 2
        ;return (T) tree[parentIndex]
        {

        } ()private void resize
        ;Object[] newTree = new Object[tree.length * 2]
        ;System.arraycopy(tree, 0, newTree, 0, tree.length)
        ;tree = newTree
        {

        } ()public int size
        ;return size
        ,
```